MCS-202	PARALLEL COMPUTING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	Define terminology commonly used in parallel computing, such as efficiency and speedup
CO2	Describe different parallel architectures, inter-connect networks, programming models, and algorithms for common operations such as matrix-vector multiplication.
CO3	Given a parallel algorithm, analyze its time complexity as a function of the problem size and number of processors.
CO4	Given a parallel algorithm, an input to it, and the number of processors, show the steps performed by that algorithm on that input.
CO5	Given a parallel code, analyze its performance, determine computational bottlenecks, and optimize the performance of the code.

MODULE I

Computational demands, advantages of parallel systems. Flynn's classification, controlled parallelism and scalability. Topologies: Mesh, binary tree, Hyper tree, Cube Connected cycles, shuffle-Connected Exchange; Uniform Memory Access (UMA & Non uniform Memory Access (NUMA) Multi processor System.

MODULE II

PARAM Model of Parallel Computation, PARAM Algorithms; Parallel Reductions, Prefix sum, List Ranking, Merging of Two Sorted List.

Mapping and Scheduling; mapping of Data from Topology to other (Ring to 2-D Mesh, Binomial trees to 2-D mesh, Rings & mesh into 2-D Mesh, Ring & Mesh into Hypercubes), Load balancing, Static scheduling on UMA multi processor systems.

MODULE III

Applications of parallel computing: Matrix Multiplication, Sorting (bitonic Merge sort, parallel quick sort, hyper quick sort), Searching a Graph (P-depth search, Breadth-Depth Search, Breath first search) , parallel Branch and bound algorithms

References

1. *Michel J. Quinn, " Parallel Computing: Theory and Practice," McGraw-Hill.*
2. *Kai Hwang, "Advanced Computer Architecture," McGraw-Hill.*

Course Outcome:

CO1	To develop an understanding of various basic concepts associated with parallel computing environments.
CO2	To understand the effects that issues of synchronization, latency and bandwidth have on the efficiency and effectiveness of parallel computing applications
CO3	To gain experience in a number of different parallel computing paradigms including memory passing, memory sharing, data-parallel and other approaches.
CO4	To earn experience in designing and testing parallel computing solutions to programming problems.
CO5	To develop improved communication and collaborative skills.